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The Role of Body Size in Shaping Insect Microbiomes Across Life Stages

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Declaration

I, Xinyu Xie, hereby declare that the work presented in this project is the result of my own efforts, with the contributions outlined below:

Data Collection: I personally collected all the data used in this study from the NCBI website (<https://www.ncbi.nlm.nih.gov>).

Data Processing and Cleaning: I was solely responsible for the processing and cleaning of the data, ensuring its accuracy and readiness for analysis.

Development of Mathematical Models: The mathematical models used in this project were developed collaboratively. I contributed to the formulation and implementation of the models, while my supervisor, Dr. Samraat Pawar, provided critical insights and guidance in refining the models and ensuring their appropriateness for the analyses presented.

I affirm that the above contributions accurately represent the division of work in this project.

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Abstract

Insects and their mutualistic relationships with diverse and ecologically significant microorganisms provide key insights into holobiont research and the broader understanding of host-microbe interactions. In this study, I utilized high-throughput 16S rRNA gene sequencing to analyze the microbial diversity and structural composition of insect gut microbiomes, aiming to uncover the relationship between gut microbiome diversity and insect size by linear model fitting. The 16S rRNA data, collected from eight different insect orders, were analyzed to calculate alpha and beta diversity based on the relative abundance of each sample, and then a linear regression model was applied to examine the relationship between insect body size and gut microbiome diversity. The results show that insect body mass is positively correlated with the diversity of its gut microflora, and the explanatory power of the model significantly increased after incorporating the factor of life stage, revealing the importance of considering both insect body size and life stage in studies of microbiomes and their hosts. This research lays a foundation for further understanding the complex interactions between insects and their gut microbiota, and provides a basis for future studies exploring the ecological and evolutionary implications of these interactions, as well as their potential applications in fields such as pest control, ecosystem management, and biotechnology.

1 Introduction

The role of microbial communities in host biology has been progressively acknowledged as essential in the pursuit of elucidating the intricate interrelationships between organisms and their environments (Al-Kahtani and A. Taha 2014, Armitage et al. 2022, Simon et al. 2019). In recent decades, an enhanced understanding of the complexity and importance of these interactions has led to the creation of new theoretical frameworks (Chisholm et al. 2006, Brown et al. 2013, Proal et al. 2017, Hu et al. 2024). Among these, the holobiont concept is outstanding for integrating the close connections between hosts and their associated microorganisms into a single entity. In the early 1990s, Lynn Margulis was the first to introduce this concept as part of her work in symbiogenesis and endocytobiology. She defined the holobiont as “the symbiotic complex” (Margulis 1990) or “the product, temporary or permanent, of the association between its constituent bionts” (Margulis 1993). Originally introduced to describe symbiotic relationships that collectively form an integrated biological entity, the holobiont has since become a vital framework for studying the interdependence between hosts and their microbial communities. This framework has provided profound insights into the anatomy (Koide 2023), metabolism (Lesser et al. 2022), development (Lyu et al. 2021), immunity (Posadas et al. 2022), and evolutionary dynamics of these composite organisms (Suárez 2020), highlighting their growing significance in ecological and evolutionary research.

The prevalence of mutualistic relationships with microorganisms, which are characterized by immense diversity and extensive ecological functions, has made insects a

focal point of holobiont research (Iltis et al. 2021). These organisms serve as an optimal paradigm for investigating the intricate interactions across diverse biological structures. Symbiotic associations between insects and microorganisms are ubiquitous and fundamental, contributing significantly to the evolutionary proliferation of this extensively varied clade. The idea that “to study an insect is to study the holobiont” reflects a growing consensus within the scientific community (Six 2013). This perspective underscores the integrative nature of insect-microbe interactions (Holt et al. 2024), highlighting their importance in understanding broader ecological and evolutionary processes (Gupta and Nair 2020). Additionally, insects’ relatively short life cycles (Kojour et al. 2020), varied diets (Gutiérrez et al. 2020), and adaptability to diverse environments make them particularly useful in experimental setups designed to study the dynamics of host-microbe interactions over time (Wilson and Fox 2020). For instance, studies on insect holobionts have revealed how microbial symbionts can influence host behaviors, such as foraging (Buehlmann et al. 2020), reproduction (Leith et al. 2021), and even resistance to pathogens (Vogel et al. 2022), demonstrating the profound impact of these relationships on the evolutionary success of insects. As holobionts, insects exemplify how host organisms and their microbiota co-evolve, adapt, and thrive together, making them indispensable in the study of microbial symbiosis and its implications for biodiversity and ecosystem function.

Given the pivotal role of insects in ecosystems and their complex interactions with symbiotic microorganisms, the intestinal microbiota stands out as particularly significant (Lu et al. 2024), revealing that it plays a key role in host nutrition, detoxification of toxic compounds, or reproduction (Muñoz-Benavent et al. 2021). Throughout evolutionary history, a symbiotic relationship has developed between insects and their gut microbes, and this interaction has unquestionably driven the evolution of insects (Armitage et al. 2022). The gut microbiome of insects not only influences digestion and absorption of food by the host (Cummings 1984), but also plays an essential role in affecting host development (Sommer and Bäckhed 2013), immunity (Pickard et al. 2017) and behavior (Pamer 2016). Moreover, some gut microbes actively degrade toxic compounds such as phytotoxins and pesticides, which contribute to the environmental adaptation of host insects (Itoh et al. 2018). In addition to these ecological and evolutionary impacts, the insect gut microbiota has potential to provide enormous bio-resources for applications in medicine and industry, as gut microbes have evolved under natural conditions to produce highly effective and optimized antimicrobial peptides and digestive enzymes (Jang and Kikuchi 2020).

The species-area relationship (SAR) is a well-established ecological principle that describes how species richness increases with habitat size (Veresoglou and Johnson 2023), as larger areas typically support more species due to greater habitat diversity and reduced extinction risks. Similarly, in insect systems, the size of the host—a factor linked to physiological and metabolic rates—may also influence the diversity and functionality of gut microbiota. This correlation suggests that the physical and biological constraints imposed by the insect’s size could shape the gut microbial ar-

chitecture in ways that parallel SAR, revealing previously unrecognized patterns of microbial assembly (Engel and Moran 2013b). In mulberry-feeding lepidopteran insects, larger species can ingest more plant material and have higher nutrient extraction efficiency, leading to greater gut microbiota diversity and higher metabolic potential compared to smaller species (Chen et al. 2019). Larger beetle species often need to digest large amounts of woody or fibrous materials, which are difficult to break down and require a more complex microbiota, a complexity less common in smaller beetles (Yang et al. 2020). In social insects like ants and bees, larger queens typically have more diverse gut microbiota, potentially linked to their longevity and reproductive capacity (Liberti and Engel 2020). Body weight serves as a reliable indicator of overall body size, it provides a comprehensive measure that accounts for both the length and volume, offering a more accurate reflection of its overall physical mass (Nijhout et al. 2006). These observations lead to a hypothesis that there might be a positive relationship between body weight and microbial diversity across insect species.

Over 80% of all described insect species are holometabolous—they undergo complete metamorphosis (Grimaldi and Engel 2005), which includes a pupal stage intercalated between the larva and the adult. During an insect's development, body weight changes dramatically, especially larval stages, where each larval instar's initial mass is a constant multiple of the previous one—a phenomenon known as Dyar's rule (Nijhout et al. 2006). The larval stage is often a critical period for nutrient intake, where rapid growth in body weight and size occurs, leading to a higher diversity and functionality of the gut microbiota to support complex digestive needs. During the pupal stage, significant metabolic changes occur, potentially leading to stabilization or reduction in body weight until further changes in the adult stage (Manthey et al. 2022). Beyond direct effects on body size and weight, factors such as metabolic demands, gut structure (Grimaldi and Engel 2005), environmental exposure (Engel and Moran 2013b), and dietary changes (Chen et al. 2019) also profoundly influence the gut microbiota communities. These impacts lead to a hypothesis that the influence of body weight on the diversity and composition of insect gut microbiota varies significantly across different life stages.

Could there be a positive relationship between body weight and insect gut microbial diversity? Does the influence of body weight on the diversity and composition of insect gut microbiota vary significantly across different life stages? In order to answer the two questions above, I aim to employ high-throughput 16S rRNA gene sequencing to dissect the microbial diversity and structural composition of insect gut microbiomes. By integrating bioinformatics tools with ecological and biological data, my research aims to deepen the understanding of the microbiome's role in insect life and its implications for broader ecological and evolutionary patterns, addressing the largely unexplored relationship between insect size and the diversity and structural composition of its gut microbiota.

2 Methods

2.1 Data compilation

The sequence data were downloaded from NCBI Taxonomy browser by conducting systematic review and search using “insect gut metagenome” as the organism. This search resulted in a total of 641 BioProject entries. Thereafter, I independently conducted an analysis of data availability, revealing that only 541 of these projects contained usable raw data. Many of these datasets were excluded for the following reasons: they were either intended for other analyses (such as ITS, metagenome DNA, or full RNA sequencing), had low raw data quality with too much noise, or were multiplexed without barcode files for demultiplexing. Additionally, some datasets had imprecise structures or metadata, or the sequence quality was significantly low, making extraction and analysis difficult. Ultimately, only 266 studies had a clear 16S rRNA sequence data structure and were fully accessible and reproducible on QIIME2 (qiime2-amplicon-2024.5). All insect life stage data were provided by metadata files from NCBI BioProject, and body mass data, represented as live mass, were obtained from various published data repositories and articles.

2.2 Data hygiene

After importing the initial fastq data into QIIME2 (Bolyen et al. 2019), the input forward and reverse read sequence files were quality controlled and trimmed using the Cutadapt tool, which included trimming bases with quality scores less than 20 from the end, discarding reads with trimmed lengths less than 150, trimming any unknown bases (N) from the end of the sequence, and finally exporting the processed forward and reverse read sequences separately to the specified file path. After initial quality control, denoising was performed using DADA2 to crop the excessively high quality parts again based on the data processed in the previous step, this step can further filter out segments of sequences that still have very low-quality scores even after the initial trimming process.

2.3 Feature annotation

In QIIME2, DADA2 is employed to identify OTUs (Operational Taxonomic Units), with denoising performed separately on forward and reverse reads, followed by the merging of OTUs at the end of the workflow. DADA2 is an open source package designed to model and correct for amplicon errors in Illumina sequencing data (Callahan et al. 2016). The algorithm implements a quality-aware model that begins by grouping all amplicon reads with identical sequences into unique sequence sets, recording the abundance and consensus quality profile for each set. These unique sequence sets are then grouped into individual partitions, with the most abundant partition (the one with the highest number of copies) designated as the center of

that partition. The similarity of each unique sequence set to the central set is calculated, and sequences with the greatest divergence from the center are identified as new partition centers. Subsequently, all unique sequence sets are redistributed between the two partitions based on their similarity to the respective centers. This iterative process continues until the sequences are partitioned in a way that aligns with the error model used for OTU creation (Callahan et al. 2016).

The DADA2 algorithm primarily relies on two criteria: the abundance of each amplicon sequence (a highly abundant sequence is likely to result from true biological variation rather than sequencing errors) and the pairwise similarity between sequences (i.e., the error rate). A classifier is then used to identify each OTU table. Each table was created for each taxonomic level (kingdom, phylum, class, order, family, and genus). Sequences that could not be assigned to the corresponding taxonomic level (but were assigned to a higher level) were accumulated into a new category labeled “unidentified.”

OTUs were clustered with 97% similarity cutoff using Vsearch and chimeras were removed using Vsearch when the proportion of non-chimeric sequences was below 75%. The phylogenetic affiliation of each 16S rRNA gene sequence was analyzed with the silva (silva-138-99-nb-classifier.qza) 16S rRNA database using confidence threshold of 70%.

2.4 Alpha and beta diversity analyses

Phylogenetic diversity analyses of core metrics were performed using the QIIME2 tool, by specifying the input feature table file (clustered-table.qza), sampling depth (2000), and metadata file (tr_manifest.tsv), calculating and comparing between-sample diversity indices, and finally output the results to the specified directory (core-metrics-results), which includes 8 data files and 2 visualisation result files. The data files cover Alpha diversity indices (e.g., Pielou Evenness, Shannon Entropy and Observed OTUs), distance matrices (e.g., Bray-Curtis distance matrices and Jaccard distance matrices), and the principal coordinate analysis (PCoA) based on these distance matrices results. The visualisation file uses the Emperor tool to visualise the PCoA results for the different distance matrices (Jaccard distances and Bray-Curtis distances). This section’s results were automatically generated using QIIME2 software, focusing on microbial diversity and community composition at the species level for insect gut microbiota, with each BioProject serving as the analytical unit. However, within each BioProject, there are often multiple experimental samples derived from one or more insect species. Although this approach effectively captures intraspecific alpha diversity, it does not facilitate interspecific beta diversity analysis. Moreover, the OTU table at the species level is overly complex and extensive, with many species not accurately annotated, leading to challenges in analysis. Therefore, for the subsequent alpha and beta diversity analyses, the microbial family level was selected from the successfully constructed OTU table, providing a more manageable

and focused perspective.

After the family-level OTU table was processed using the *vegan* package (version 6.1) in R, alpha diversity analyses yielded four key metrics: Richness, Pielou's Evenness, Shannon Index, and Simpson Index. Following the completion of the alpha diversity analysis, I proceeded with the beta diversity analysis utilizing the Bray-Curtis distance measure. The first step involved using the *vegan* package to compute the Bray-Curtis distance matrix, which captures the differences in microbial community composition among the samples. Next, Principal Coordinates Analysis (PCoA) was performed on the distance matrix to reduce dimensionality, thus projecting the multidimensional differences between samples into a lower-dimensional space. This visualisation allows for an intuitive display of differences in the structure of the microbial community among samples. PCoA facilitates the identification and visualization of community structure differences across various experimental conditions or groups, further elucidating potential ecological or functional relationships. This analytical approach provides critical insights into the variability of microbial communities under different environmental or experimental conditions.

2.5 Statistical analyses

To investigate the impact of insect body weight on gut microbial diversity, this study employed multiple linear models. The primary response variables included the Shannon diversity index and log-transformed family-level richness, with log-transformed body weight (LogMass) as the key predictor. Additionally, given the potential influence of different life stages on microbial diversity, life stage (larva, pupa, adult, and egg) was introduced as a categorical variable. During the model construction process, I first examined the direct effect of body weight on diversity, then incorporated life stage into the linear model to explore whether it moderated the relationship between body weight and diversity. The model fit was initially evaluated using ANOVA and p-values to assess the significance of the explanatory variables. Subsequently, AIC, BIC, and adjusted R-squared values were utilized to determine the best combination of explanatory variables for model comparison, ensuring that the selected model provides the most parsimonious fit while balancing goodness-of-fit and model complexity. Moreover, standardised regression coefficients were calculated to compare the relative contributions of body weight and life stage in predicting microbial diversity.

3 Results

3.1 Summary of the 16s rRNA sequencing data

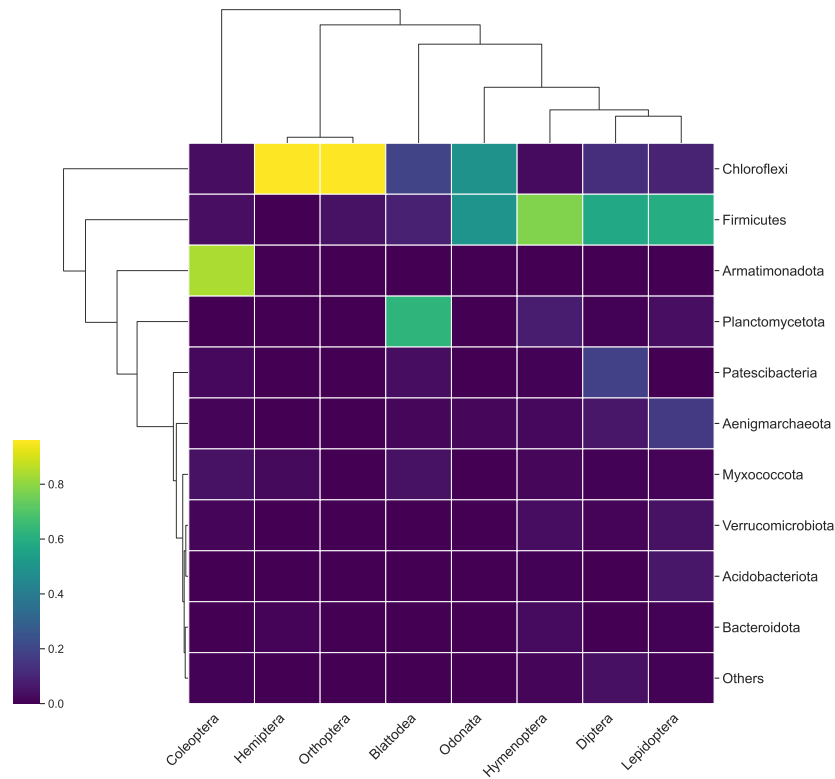
In total, 116156 OTUs were detected at a 97% similarity level. For further analysis, this study focuses on the microbial family level, including 61 phylum-level OTUs, 166 class-level OTUs, 385 order-level OTUs, and 670 family-level OTUs.

After data processing, 9296 insect samples were obtained. Following the removal of insect species with a frequency of less than 0.1% of total, 8113 samples remained, representing 105 species from 8 different insect orders: Blattodea, Coleoptera, Diptera, Hemiptera, Hymenoptera, Lepidoptera, Odonata and Orthoptera. There were significant differences in the sample size and relative proportion among these orders. The Hymenoptera order had the largest number of samples, with 4180, accounting for 51.53% of the total samples. In contrast, the Diptera order also showed a notable presence with 1302 samples, comprising 16.05% of the total. Hymenoptera and Diptera dominated the sample set, while orders like Odonata and Orthoptera had relatively fewer samples, making up 0.60% and 3.55% respectively. These differences in sample size and proportion may reflect the representativeness and specificity of microbial communities within different insect orders in gut microbiome studies.

3.2 Composition in gut microbiota diversity by different insect order and life stage

In the analyzed samples, notable differences were observed in the relative abundance of bacterial phyla and families across different insect orders. Chloroflexi and Firmicutes exhibited unique distribution patterns, with Chloroflexi being dominant in Hemiptera, Odonata, and Orthoptera (0.958, 0.485, and 0.956, respectively) but significantly lower in Coleoptera and Hymenoptera (0.032 and 0.027). Conversely, Firmicutes peaked in Hymenoptera at 0.778 but was extremely rare in Hemiptera at 0.004. At the family level, KD4-96, within the Chloroflexi phylum, showed considerable abundance in Diptera (0.299) and was also prominent in Odonata (0.230) and Orthoptera (0.103). Anaerolineaceae was more abundant in Hemiptera (0.046), while Christensenellaceae was most prominent in Blattodea (0.088) and appeared in Lepidoptera (0.016).

(A)



(B)

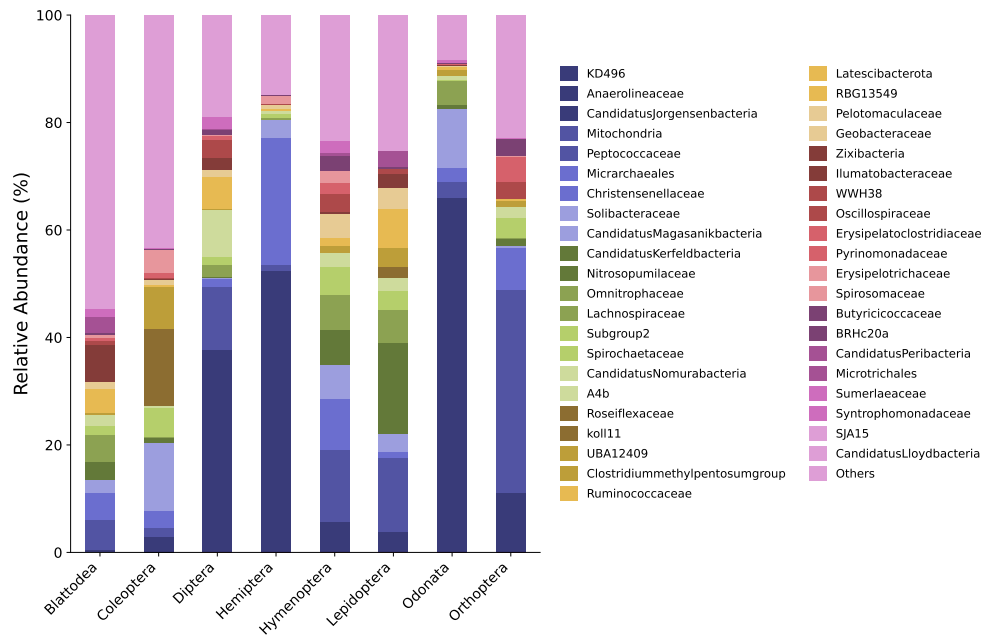


Figure 1: Across insect orders, the heatmap (A) shows the Top 10 microbial phyla, dominated by Chloroflexi and Firmicutes, with a dendrogram illustrating the relationships between samples. The stacked bar chart (B) presents the Top 50 microbial families, including KD4-96, Anaerolineaceae, and Christensenellaceae. Groups with a relative abundance $\leq 0.1\%$ and uncultured taxa were grouped as “Others”.

The Principal Coordinates Analysis (PCoA) of insect community data illustrates clear clustering patterns among insect orders, showcasing differences in microbial community composition. Overall, the eight insect orders show a significant degree of overlap, indicating similarities in their microbial communities. Notably, Coleoptera forms a compact and distinct cluster, signifying consistent microbial communities. Conversely, Diptera spans almost all sample points, reflecting the most diverse microbial composition. The plot's axes, PC1 and PC2, account for a small but notable proportion of the variance in the data, with PC1 explaining 7.38% and PC2 explaining 4.72%.

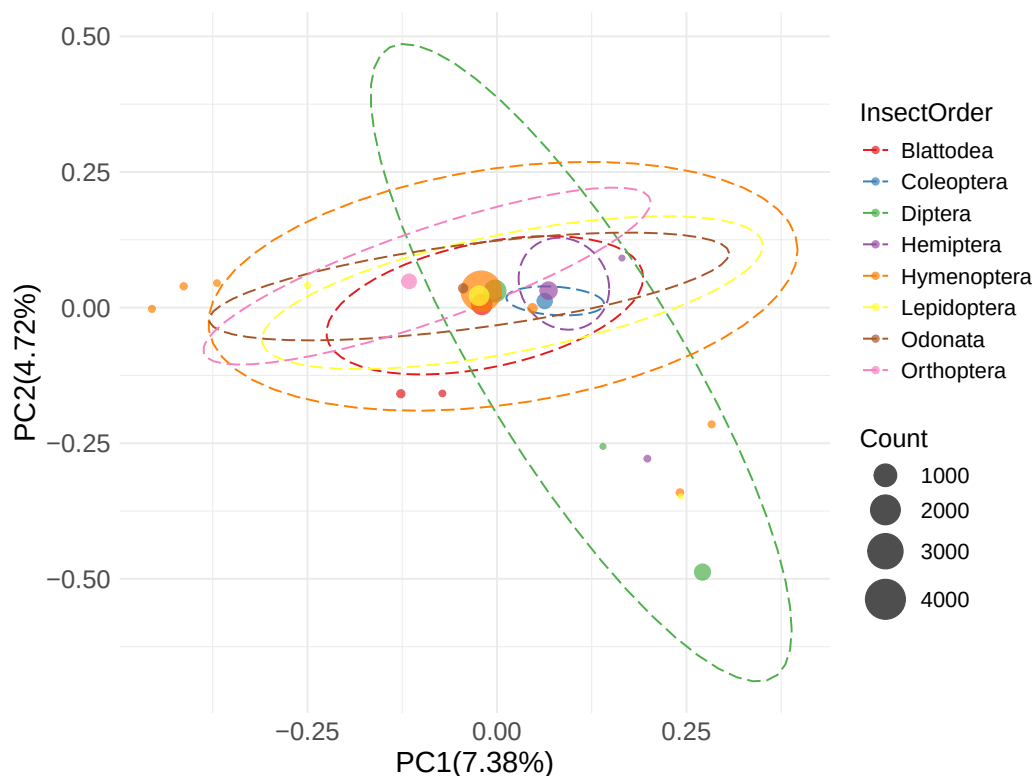


Figure 2: The Principal Coordinates Analysis (PCoA) of insect community data reveals differences in microbial community composition among insect orders, with Coleoptera forming a tight cluster, Hymenoptera showing a wide distribution, and Diptera exhibiting the highest microbial diversity.

A total of 2,526 samples with lifestage information were available, accounting for 31.14% of the total samples. Further considering the differences in gut microbiota composition across different life stages of various insects, the PCoA results show that the data for the egg and larva stages are relatively dispersed, particularly for the egg stage, which covers a wide range, indicating greater variability in microbial community structure during these stages. In contrast, the pupa stage exhibits the most tightly clustered data, reflecting less variability in the microbial community structure. The adult stage's sample points are distributed between the two, with the confidence ellipse showing a moderate coverage range.

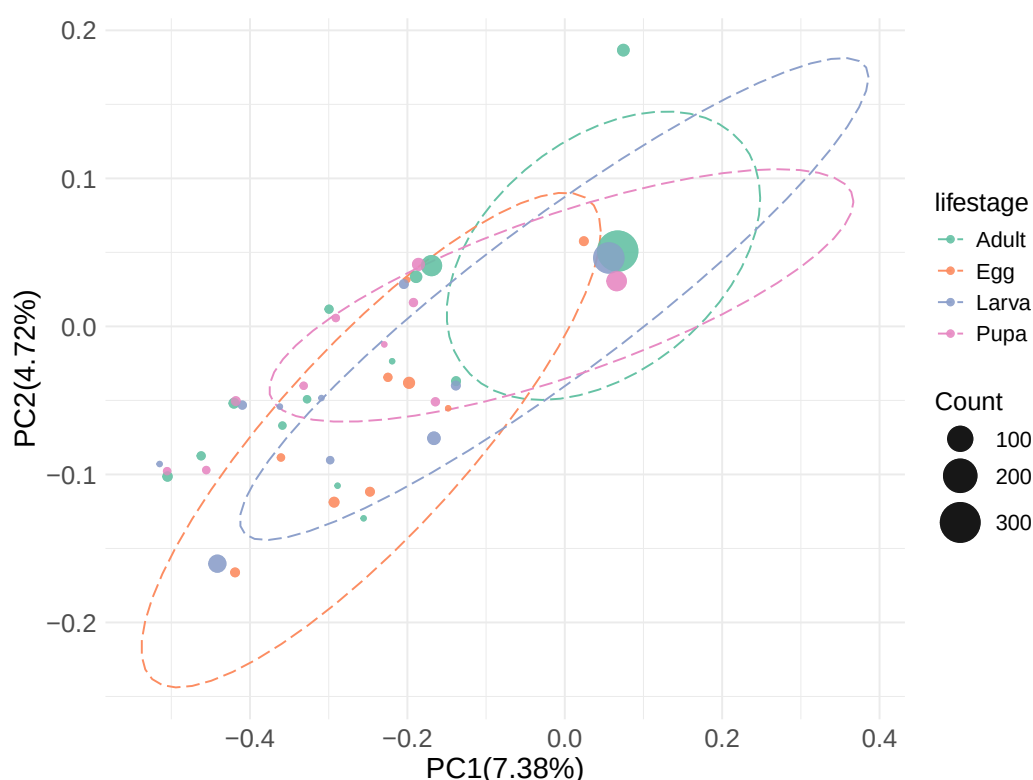


Figure 3: Principal Coordinates Analysis (PCoA) of Microbial Community across Insect Life Stages, the microbial community variability is highest during the egg and larva stages, lowest during the pupa stage, and intermediate during the adult stage.

3.3 Relationships Between Diversity, Body Weight, and Life Stage

Since different insects show similarities in microbial composition, further diversity analyses are particularly important. Among the investigated insect orders, substantial variation was observed across the different alpha diversity indices (see Supplementary Table 2). Blattodea exhibited with the highest diversity in all metrics, including the top Shannon index (2.80), Simpson index (0.85), Richness (78.06), and Evenness (0.65), indicating the most diverse and evenly distributed microbial community and Coleoptera was close behind. On the contrary, Diptera and Hemiptera

showed lower diversity values, with Shannon indices of 0.98 and 0.81 respectively, Simpson indices of 0.42 and 0.37, and the lowest Richness among all orders. In most instances, these four indices were strongly positively correlated. Notable exceptions were found in Hymenoptera and Orthoptera. Hymenoptera had a relatively low Shannon index of 1.15 and low Richness, but a moderate Pielou's Evenness index of 0.46, while Orthoptera implied high diversity and evenness despite its lower species richness compared to some other orders.

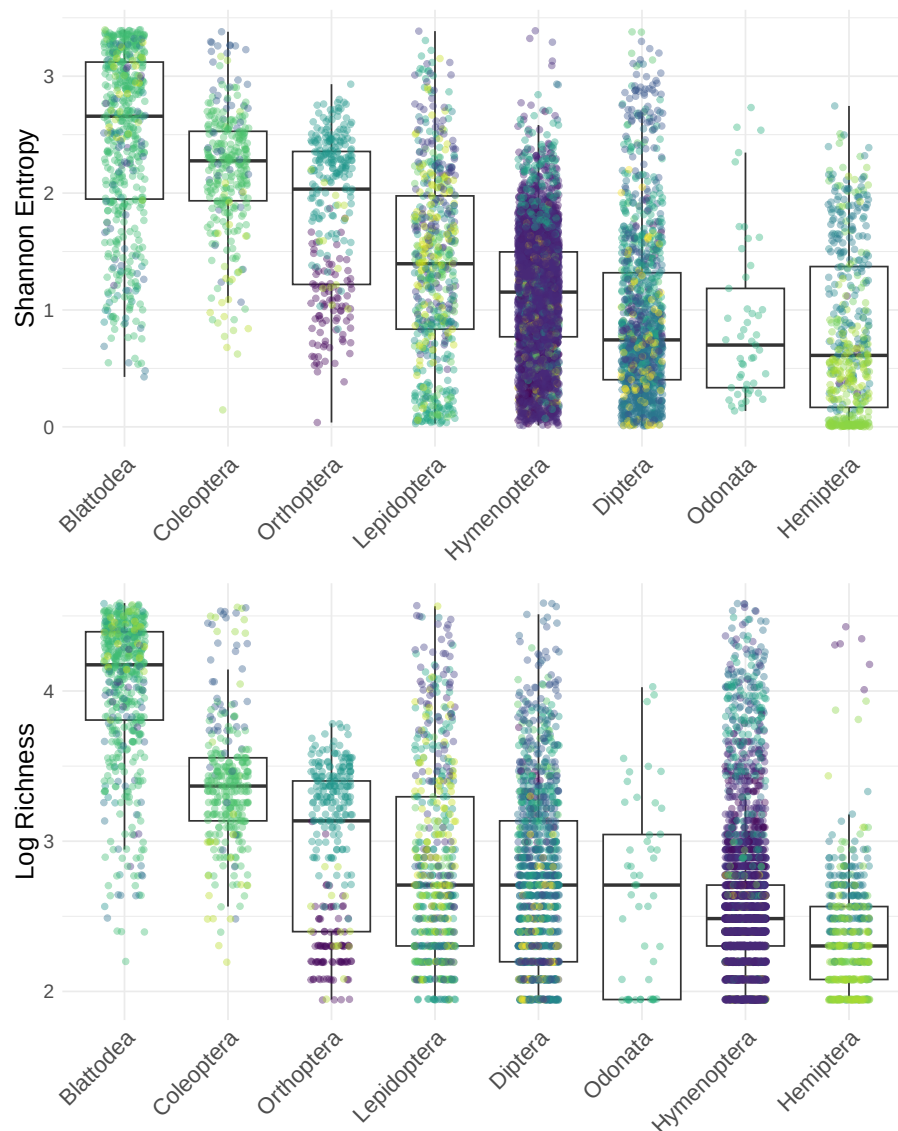


Figure 4: Alpha Diversity Indices across Different Insect Orders

Lifestage, as an important classification criterion, was also considered in the analysis. Based on this subset of data, further diversity analyses were conducted. From the perspectives of Shannon diversity and richness, a decreasing trend is observed across the different lifestages of insects, following the order larva-pupa-adult-egg.

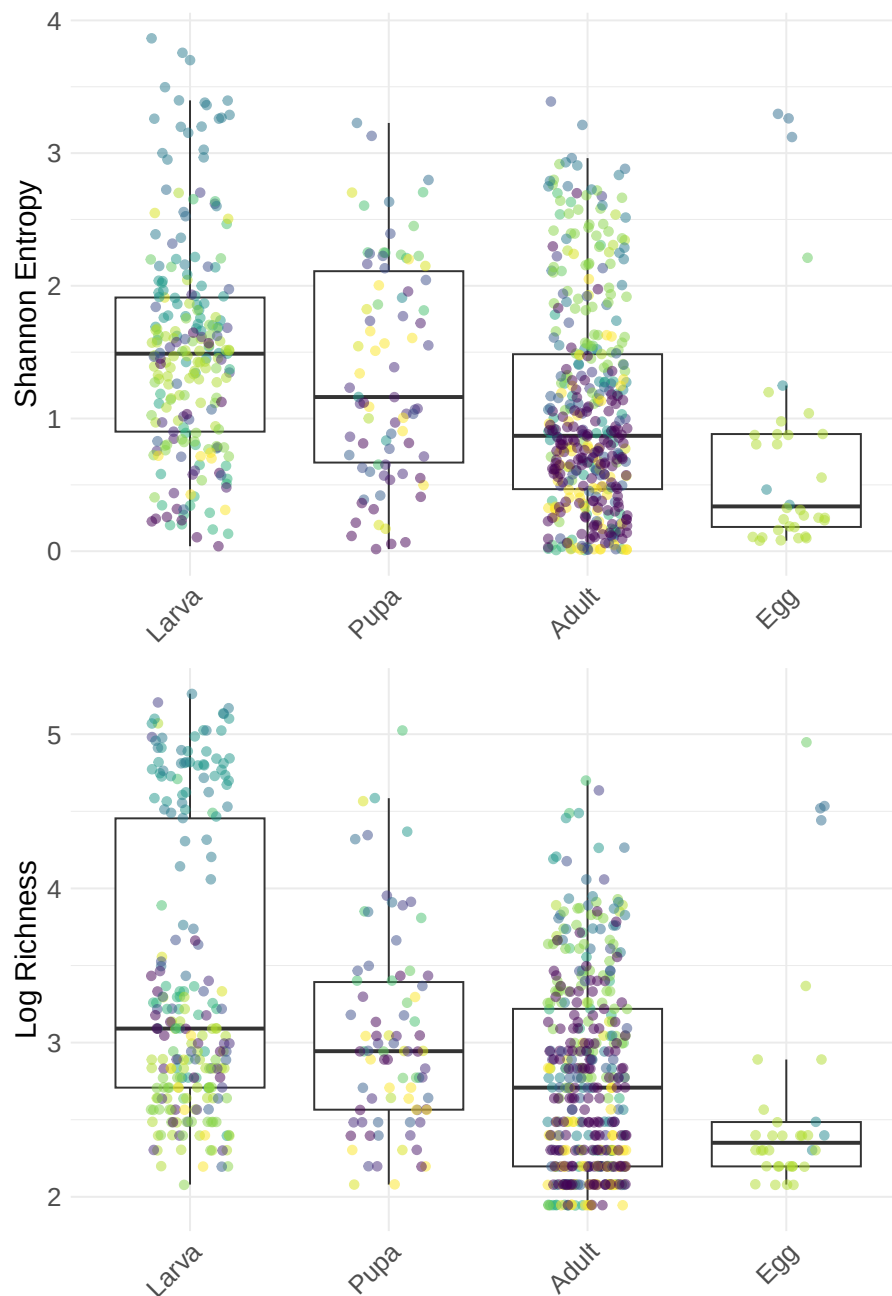


Figure 5: Diversity Patterns across Life Stages

In examining the connection between insect body weight at and diversity (Shannon index and species richness) as shown in Table 1, it was found that body weight significantly influences both metrics. Following an F-test, p-values for both models were under 0.01, while the R^2 value had a obvious difference, with body weight accounting for 21.62% and 1.7% of the variance respectively. This reveals an evident increase in diversity as insect body weight increases. However, model fit varies based on the diversity index used. When Shannon entropy was the dependent variable, Model 1 outperformed Model 3, where LogRichness was the dependent variable. Model 1 had lower AIC and BIC values (1686 and 1700, respectively) compared to Model 3 (1762 and 1776, respectively). This suggests that the Shannon index offers a better model fit and greater explanatory power. Standardized regression coefficients reveal that body weight had a stronger impact on the Shannon index (0.429) than on species richness (0.012). Thus, the choice of diversity index is crucial for the model's explanatory power and fit.

Table 1: Model Fit Metrics and Comparison Coefficients

Model	AIC	BIC	Adjusted R^2
Model 1 (Shannon $\sim$ LogMass)	1686	1700	0.22
Model 2 (Shannon $\sim$ LogMass + lifestage)	1676	1703	0.23
Model 3 (LogRichness $\sim$ LogMass)	1762	1776	0.017
Model 4 (LogRichness $\sim$ LogMass + lifestage)	1676	1704	0.126

Model	F-value	p-value	S.C. (LogMass)	S.C. (lifestage)
Model 1	5.44	0.002***	0.429	NA
Model 2	-	-	0.429	-0.019 (P), -0.145 (A), -0.034 (E)
Model 3	32.25	<2.2e-16***	0.012	NA
Model 4	-	-	0.012	-0.140 (P), -0.390 (A), -0.197 (E)

To investigate the potential influence of life stage on insect gut microbial diversity, life stage was incorporated as a moderating variable in the models. This resulted in variations in the relationship between body weight and diversity. Models including life stage (Model 2 and Model 4) showed improved fit compared to those considering only body weight (Model 1 and Model 3), with decreased AIC and BIC values. The explanatory power of body weight varied with the diversity index used. In the Shannon entropy model, life stage slightly increased the explanatory power of body weight (Adjusted R^2 from 0.216 to 0.230), whereas in the richness model, it increased (Adjusted R^2 from 0.126 to 0.017). This indicates that life stage moderates the relationship between body weight and diversity, potentially affecting insect physiological status and ecological behavior.

While models combining both body weight and life stage improve the prediction of gut microbial diversity compared to body weight alone, this suggests that life stage as an overall variable considerably impacts the model. It does not necessarily mean that life stage influences diversity more than body weight. To evaluate the greater

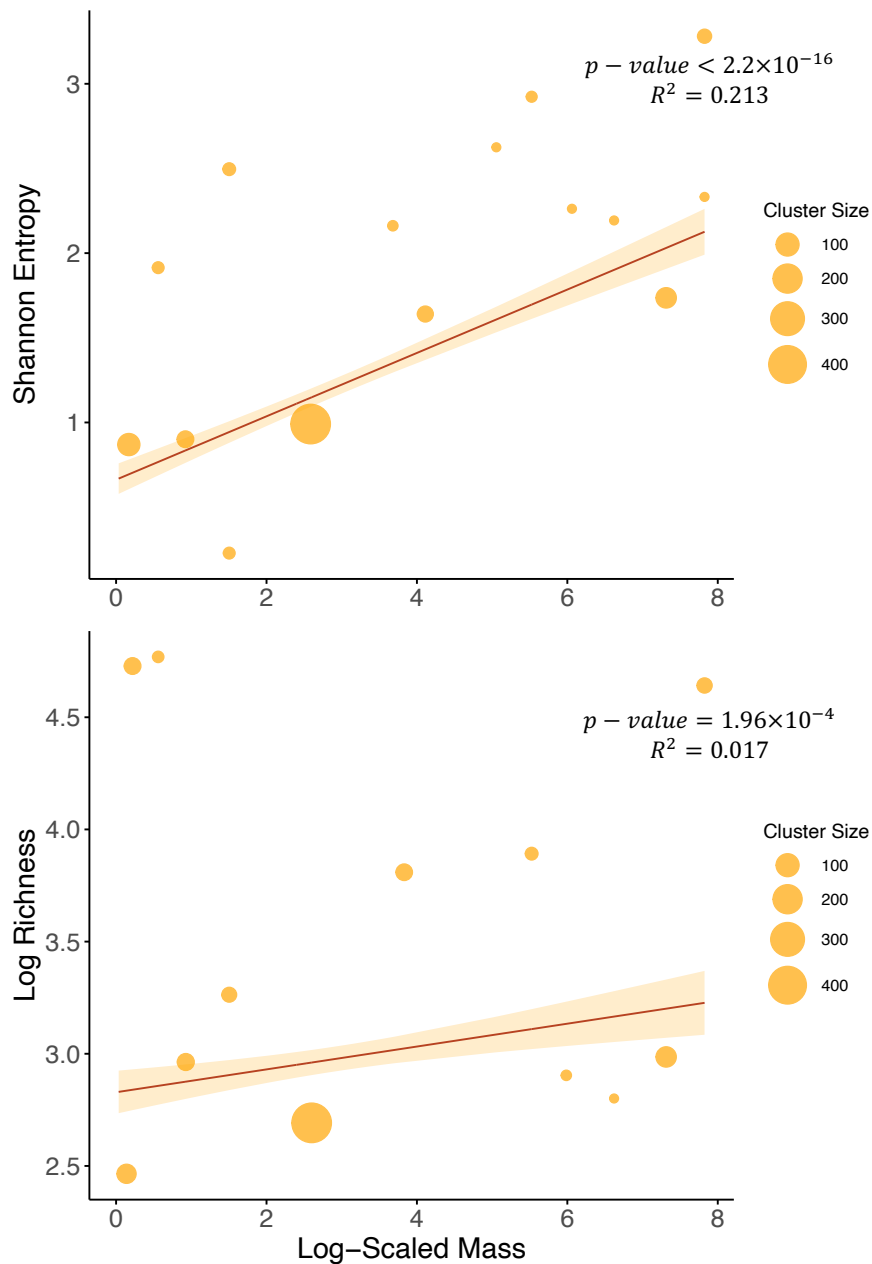


Figure 6: This figure presents the regression analysis of Shannon Index and Log Richness against Log-scaled mass, where both richness and mass were log-transformed using $\ln(x + 1)$, effectively avoiding negative values and achieving standardization. The results indicate a significant positive correlation between Shannon Index, Log Richness, and Log-scaled Mass.

influence between body weight and life stage, standardized regression coefficients were compared. Results showed manifest variation depending on the diversity index. For the Shannon index, the standardized coefficient of body weight was 0.429, significantly higher than in the species richness model (0.012), indicating a more

substantial impact on the Shannon index.

Comparing models assessing the impact of insect body weight and life stage on diversity revealed different sensitivities based on the diversity index. In the Shannon index model, body weight and life stage together explained 22.99% of the diversity variance (Adjusted $R^2 = 0.23$), with body weight positively impacting diversity (standardized coefficient = 0.429). In contrast, the species richness model had lower explanatory power (Adjusted $R^2 = 0.13$), with body weight having a smaller impact (standardized coefficient = 0.012). The results indicate that as insects transition from the larval stage to other life stages, the standardized regression coefficients are consistently negative, suggesting a progressive decrease in gut microbial species diversity. The adult stage shows the greatest reduction in diversity, with a standardized regression coefficient of -0.145 in the Shannon index model and -0.390 in the species richness model. These findings suggest that life stage more profoundly affects species richness, while the Shannon index is more sensitive to body weight's impact on community structure, highlighting the importance of selecting the appropriate diversity index for accurately assessing the body weight-diversity relationship.

4 Discussion

In this study, I used 16S rRNA analysis to calculate the gut microbial diversity and composition of insects of various body sizes, and employed linear models to investigate the relationship between insect body mass and gut microbial diversity in order to understand how body size influences microbial communities across different life stages. Based on the results, there is a significant positive correlation between insect body mass and gut microbial diversity, which supports the initial hypothesis that larger insect body mass is associated with higher microbial diversity. When life stage is introduced as a moderating variable, it significantly influences the relationship between body mass and gut microbial diversity, particularly with a pronounced negative effect during the adult stage on species richness. This finding supports my second hypothesis that life stage plays a crucial role in modulating the relationship between body mass and microbial diversity, with this influence manifesting differently across various diversity metrics.

The strong positive correlation between body size and microbial diversity not only confirms the close link between larger body mass and microbial diversity but also implies a deeper cause-and-effect relationship between the two: body mass might directly boost gut microbial diversity through different physiological mechanisms, including changes in insect niche and habitat (Brasil et al. 2020), dietary habits (Gómez-Corona and Valentin 2023), and predatory behavior (Silva et al. 2020). The adult insect's body size is significantly influenced by the environmental factors it is exposed to during its growth and development. Insect species that face high environmental variability across their geographical range may exhibit a wider range of body sizes compared to more specialized species. For instance, the body size of European looper moths increases with a broader latitudinal range (Seifert et al. 2022), potentially affecting the composition and diversity of their gut microbiota. Additionally, dietary habits are closely related to insect body weight, which in turn impacts gut microbial diversity. Larger herbivorous insects, such as large beetles (Jing et al. 2020), often consume large amounts of plant cellulose. In woody materials, the cellulose in plant cell walls is protected by lignin, which serves as a barrier to carbohydrate breakdown for wood-feeding insects (Engel and Moran 2013a). Since insects lack lignase enzymes (Douglas 2013), these indigestible substances must be degraded by complex gut microbial communities. This dietary necessity directly supports the diversity and specialization of their gut microbiota. Conversely, cockroaches, known for their omnivorous diet and relatively large body size, have a highly varied gut microbiome. This diversity is thought to be driven by their varied diet and significant food intake, which supports a wide array of microbial species that assist in digestion, nutrient absorption, and detoxification (Zhu et al. 2022). Omnivorous insects, because of their diverse diet, need gut microbes that can process multiple food sources, likely sustaining a more complex microbial community. The complexity of these microbial communities is not only present in omnivorous insects with diverse diets but also exhibits unique characteristics in predatory insects. Due to their predatory behavior, larger predatory insects may indirectly influence their gut microbial diversity by ingesting the gut microbiota of their prey. The phenomenon of a highly

transient and prey-influenced gut microbiome has been observed in the grey house spider *Badumna longinqua*. After cricket consumption, numerous cricket-derived microbes appeared in the spider's gut, resulting in a rapid homogenization of microbial communities among spiders (Kennedy et al. 2020). An increase in body size directly drives gut microbiota diversity by altering ecological niches, dietary habits, and predatory behavior.

The inclusion of life stages in the model significantly improved its explanatory power, allowing body weight and life stages to jointly account for a greater proportion of the variance in diversity. This suggests that life stages directly influence gut microbial diversity by affecting insect body size and other physiological mechanisms. However, the trend in body weight change throughout the life cycle varies among different insects. For example, some Lepidoptera insects, such as *Danaus plexippus* and *Heliconius charithonia* (Fescemyer and Hammond 1988, Boggs 2009, Wiklund and Gotthard 2005), have a greater body weight in the larval stage than in the adult stage. This may lead to a higher nutritional intake during the larval stage, resulting in a richer and more diverse gut microbiome. However, as they transition into the pupal and adult stages, the decrease in body weight, along with changes in diet and metabolic needs, may lead to alterations in the diversity and composition of their gut microbiota. In contrast, other insects, such as those in the orders Orthoptera and Coleoptera, experience a gradual increase in body weight throughout their growth process, resulting in variations in gut microbial diversity across different life stages. Additionally, changes in ecological niches throughout the insect life cycle are key factors influencing gut microbial diversity. For instance, the life cycle of dragonflies includes both an aquatic larval stage and a terrestrial adult stage (Stoks and Córdoba-Aguilar 2012), exposing the insects to vastly different microbial environments, which leads to distinct differences in their gut microbiota across life stages. In social insects such as bees, the rapid contact with the hive environment and older bees after reaching adulthood may significantly contribute to the notable increase in gut microbiome diversity. The insect gut undergoes significant changes across different life stages, affecting the diversity and composition of the gut microbiome. The basic structure of the insect alimentary canal consists of three regions: foregut, midgut, and hindgut (Chapman 1985). During metamorphosis, the exoskeleton lining of the foregut and hindgut is continuously shed with each molt, leading to a complete or near-complete disruption of colonized microbial populations. Similarly, the midgut sheds its peritrophic matrix during molting, disrupting the associated microbial populations. The remodeling of the insect gut during metamorphosis results in the complete elimination of larval gut components and bacteria, allowing the colonization of new microbes in the gut of newly emerged adults. Molting at each developmental stage provides an unstable environment for microbial colonization until the insect reaches the adult stage (Shamjana et al. 2024). The insect life stage profoundly influences gut microbial diversity and composition by affecting factors such as body weight, ecological niche, and gut structure. Understanding the roles of these factors at different life stages is crucial for uncovering the formation and evolution of insect gut microbiomes.

Hymenoptera shows low diversity and species richness with a homogeneous microbial community, while Orthoptera maintains balanced microbial diversity and evenness despite having lower species richness. From this observation, I selected the Shannon index and Richness as the main metrics for evaluating insect gut microbial diversity. This decision is based on the different sensitivities and uses of these metrics in describing microbial ecosystems. Firstly, the Shannon index considers both the richness and evenness of microbial communities, making it a more comprehensive gauge of the effect of body weight on gut microbial diversity. My findings show a significant positive correlation between body weight and the Shannon index, indicating that higher body weight is linked with greater microbial diversity. The advantage of the Shannon index lies in its responsiveness to changes in overall community structure, making it a better metric for explaining the overall impact of body weight on diversity. On the other hand, Richness, as a direct measure of diversity, primarily reflects the number of species in a community without considering their relative abundance. The results demonstrated that while the effect of body weight on Richness is smaller, Richness is more sensitive to changes in life stages. An increase in body weight can lead to the expansion of certain dominant microbial groups rather than the addition of new species, resulting in a lesser impact on species richness. Notably, the negative effect of the adult stage on Richness is significant. This implies that Richness can more effectively illustrate the specific effects of different life stages on gut microbial communities, especially when comparing the influence of various life stages on the number of species present. The choice of diversity indices can lead to different conclusions. By taking into account microbial community evenness, the Shannon index provides a more overall understanding of the impact of body weight. In contrast, species richness, by reflecting changes in the number of species, better captures the effect of life stages on microbial diversity. Using both indices broadens and deepens the interpretation of the factors influencing insect gut microbial diversity. Related to the Species-Area Relationship (SRA) (Veresoglou and Johnson 2023), both species richness and Shannon diversity increase with habitat size because larger habitats support more species due to greater niche availability, decreased extinction risks, higher habitat heterogeneity, and reduced edge effects. While species richness directly increases with the number of species present, Shannon diversity takes into account the distribution of individuals among these species, which also tends to improve with habitat size due to the same ecological principles. Therefore, both metrics usually show a positive relationship with habitat size, but Shannon diversity offers a more detailed view by incorporating species evenness.

My study has several limitations that need to be acknowledged. Firstly, the data was too heterogeneous and lacked precision, which may have impacted the accuracy of the results. The data on life stages did not encompass all samples, and the chosen data set had a wide range, leading to significant variability between different insect orders. Focusing solely on body size and life stage as indicators of gut microbiota diversity is too narrow and may overlook other potentially more significant factors, such as food source, predation methods, and living environment. It is also difficult to distinguish between the effects of life stage on body weight and its subsequent

impact on insect gut microbiota, and the direct effects of life stage on insect gut microbiota. Additionally, there are numerous other factors that determine insect gut microbiota, which further reduces the explanatory power of the generated model. In addition to the physical differences among insects, other aspects such as lifestyle should also be considered. For example, is the insect parasitic or independent? Is it a social insect or a solitary one? These factors could greatly influence the composition and diversity of gut microbiota. Social insects like bees and ants, due to their larger body size and complex social structures, typically develop intricate immune systems to maintain highly specific and diverse gut microbiota (Sheehan et al. 2020). For instance, honeybee (*Apis mellifera*) colonies raise their internal temperature as a colony-level response to prevent infections from common heat-sensitive pathogens (Starks et al. 2000). This complex colony-level immune strategy allows social insects to better withstand environmental stressors, thereby maintaining the stability of their gut microbial diversity. The group interactions of social insects may also influence the composition and diversity of their gut microbiota through close contact, shared food resources, and environmental conditions. These characteristics of group living can lead to the transmission of microbes between individuals, thereby increasing gut microbial diversity. For example, social insects like bees and ants engage in food exchange and share nesting spaces, which not only promotes microbial transmission among individuals but also stabilizes the gut microbiota diversity within the colony. For instance, previous studies have reported that the gut microbiota of honeybees is primarily acquired through social contact (Powell et al. 2014). Additionally, it is important to consider cases where body weight does not correlate with gut microbiota diversity. Termites, which evolved from cockroach ancestors, have relatively small body sizes but possess highly complex gut microbiota. The termite gut contains many specialized compartments, including several hindgut regions or paunches, which house dense and distinct microbial communities that vary significantly between different gut regions (Köhler et al. 2012).

Although my study has certain data and methodological limitations, it also highlights areas that warrant further exploration. Future research should consider a broader range of factors, such as insect lifestyle, ecological environment and those uniqueness like social insects and termites, to fully understand the diversity and evolution of insect gut microbiota.

5 Conclusion

In my research, body weight was found to positively correlate with microbial diversity, while changes in life stages significantly impact species richness. These findings indicate that insect gut microbial communities undergo significant restructuring across different life stages due to variations in body weight, ecological niches, and gut physiology. This not only provides important insights into host-microbe interactions and insect adaptation strategies but also holds potential significance for broader ecological and microbiome research.

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Data Availability Statement

Workflow developed and original scripts to replicate the study are available on GitHub: <https://github.com/Yurixxy/InsectgutMicro>

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Supplementary Information

Table 2: Diversity Metrics across Insect Order

Insect Order	Shannon Index	Simpson Index	Richness	Pielou Evenness	Proportion
Blattodea	2.80 ± 0.81	0.85 ± 0.15	78.06 ± 45.94	0.65 ± 0.15	10%
Coleoptera	2.29 ± 0.62	0.80 ± 0.14	40.07 ± 33.68	0.66 ± 0.12	4%
Diptera	0.98 ± 0.77	0.42 ± 0.27	20.87 ± 23.10	0.34 ± 0.21	16%
Hemiptera	0.81 ± 0.71	0.37 ± 0.30	12.52 ± 17.88	0.33 ± 0.27	6%
Hymenoptera	1.15 ± 0.56	0.54 ± 0.23	14.86 ± 16.38	0.46 ± 0.19	52%
Lepidoptera	1.42 ± 0.82	0.57 ± 0.28	23.51 ± 23.90	0.48 ± 0.23	8%
Odonata	0.89 ± 0.70	0.35 ± 0.27	16.71 ± 12.36	0.33 ± 0.22	1%
Orthoptera	1.83 ± 0.65	0.71 ± 0.19	20.69 ± 10.06	0.62 ± 0.15	4%